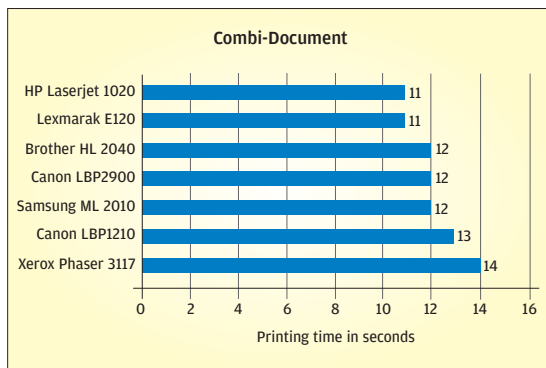




capacities ranging between one and 10 sheets.

Brother's HL-2040 and Canon's LBP-1210 were the only two printers with both USB and parallel ports. While some may think of this as backward compatibility, others may think of them as older-generation printers! All the others sport the USB 2.0 interface.

The Lexmark E-120 was the only printer in this group to offer controls in the form of Cancel and Ready buttons. The Brother HL-2040 has one large button for Ready, and the Samsung ML-2010 has one for saving toner. Canon's LBP-2900 has a button to resume printing if the printer runs out of paper. The HP Laserjet 1020 has no buttons or controls on the body.

As indicators go, the Brother HL-2040 has four, one each for toner, drum, paper, and Ready—quite useful. The rest have two indicators—Ready and Error. Canon's LBP-1210 has just one, which will not help you while troubleshooting.

Generally, printers don't come with data cables, but this time, almost all of them came with a USB cable in the package. The Quick Setup guides are comprehensive, and it's a no-brainer when it comes to installing any of these printers—all drivers installed smoothly, and Windows recognised them all without problems. Brother has a nice flash animation that shows how to install a new printer—useful for a first-time laser consumer. For open source fans, Brother is the only one that shows them some love by providing links to Linux drivers.

HP and Canon rule the roost as software goes. Canon's status window is fabulous, and provides a good amount of detail—when printing multiple pages, it

shows the progress, the page number of the current job, etc. Lexmark's software is also quite informative, but their one-size-fits-all approach might leave a layman confused—they seem to have developed generic software, which displays features that aren't even present on certain models.

Additional features, such as toner saving, sleep mode, water-marking, etc., are available with almost all the printers. Duplexing (printing on both sides) is a great way of saving paper, but none of these printers support automatic duplexing, so you'll have to manually turn pages over to save a few trees.

We didn't do a cost-per-page test, but we did note the claimed figures for the toner cartridges, and calculated the claimed cost per page. None of these lasers, even on paper, can manage to



print a page for less than a rupee—real world costs will always be higher. The Brother HL-2040 beats every printer here, thanks to its relative high output and low cartridge price. Canon's LBP-2900 was the most expensive, and each print will cost you about Rs 2. All the others work out to about Rs 1.50 per page.

Laser printers are known to be noisy, and the oldest one we have, an HP LaserJet 6L Gold, sounds like a Gattling gun when it's printing. But the times have changed! All of these printers ran quietly, with only the paper mechanism breaking the silence once in a while.

Performance

A printer's performance is measured in terms of speed and quality. Generally, there isn't much variations in terms of speed, but quality does differ wildly.

PPM

Manufacturers use PPM (Pages Per Minute) to display speeds. All the years we have been testing printers, no printer has ever lived up to its claimed ppm speed, and we were eager to see if that elusive achievement would be met this time.

Three printers claimed 20 ppm: the Brother, Lexmark, and the Samsung. The Lexmark E-120 came the closest to the claimed speed by delivering 19 ppm—a superb achievement! This machine is fast, and delivered the first (usually the slowest) print within 9 seconds, breaking all earlier records. The Samsung and Brother were fast too, delivering 18 ppm.

The Xerox Phaser 3117 promised to print 17 ppm and delivered 15, whereas the HP Laserjet 1020 delivered one less than the 15 ppm it claimed. The two Canons were the slowest—the LBP-2900 managed only 11 ppm, and the LBP-1210 did a little better with 13 ppm. Overall, this was still a good show by all the printers, and this shows that not only are printer engines speeding up, manufacturers are also being more realistic with their claims. Of more significance to us was the fact that all the printers were able to break the 12-second barrier for the first printout, which is generally the most important aspect of printing.

The Litmus Test

Our Point-Size document is a simple document with one sentence in varying fonts. Lexmark's E120 set the pace by delivering a printout in less than 9 seconds—a new record. The HP, Brother, and Canon LBP-1210 tied for second place with 10 seconds. The Samsung and Xerox had the highest spool times, and were therefore the slowest for the first print.

The Acid Test

Our combi-document is a complex document that all printers hate. It is a 35 MB file that contains different types of text,